Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			**Indicative content:** **Processes that add CO_2:** 1. The role of micro-organisms when living things die and decay. These release carbon dioxide. 2. Animals and plants release carbon dioxide during respiration. 3. Volcanic activity **Processes that remove CO_2:** 1. Carbon was stored during formation of fossil fuels 2. Carbon dioxide is taken up by green plants in photosynthesis 3. Seas and oceans absorb carbon dioxide 4. Carbonate compounds locked in rocks **Stabilised conditions:** 1. Presence of carbon dioxide caused a greenhouse effect 2. so keeping it warmer than it would otherwise be (is important to stabilize conditions for life) 3. lack of human activity meant conditions were stable	6			6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				5-6 marks At least 5 points from all sections *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3-4 marks** At least 3 points from at least 2 sections *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1-2 marks** 1/2 points from any section. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*						

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Values for 30 km are: car 1.8 or train 0.54 (1) Difference = 1.26 (1) 250 × 1.26 = 315 (1) 10 journeys so saving = 3150 (kg) (1) **OR** Car: 1.8 × 250 = 450 (1) Train: 0.54 × 250 = 135 (1) 450 - 135 = 315 (1) 315 × 10 = 3150 (1) **OR** Car: 1.8 × 250 = 450 (1) Train: 0.54 × 250 = 135 (1) 450 × 10 = 4500 and 135 × 10 = 1350 (1) 4500 – 1350 = 3150 (1) Ecf on subtractions only 630 / 315 / 1575 on answer line (3) 1350 / 4500 on answer line (2)		4		4	4	
				Question 5 total	**6**	**4**	**0**	**10**	**4**	**0**